

# SGA 6: Exposé VIII

The Riemann–Roch theorem, statement and preliminaries

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## Exposé VIII. The Riemann–Roch theorem

by Pierre Berthelot

### 1. Complete-intersection morphisms

**Definition 1.1.** Let  $f : X \rightarrow Y$  be a morphism of schemes. One says that  $f$  is a *complete-intersection morphism* if, for every point  $x$  of  $X$ , there exist an open neighbourhood  $U$  of  $x$  and a smooth  $Y$ -scheme  $V$  such that the restriction of  $f$  to  $U$  factors as

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow f & \downarrow g \\ & & Y, \end{array}$$

where  $i$  is a regular immersion (VII 1.4). After replacing  $V$  by an open subset, one may suppose that  $i$  is a regular closed immersion.

In this definition one may replace  $V$  by a vector bundle  $Y[T_1, \dots, T_n]$ . Indeed, since the notion is local, one may suppose that  $V$  is affine and that its image in  $Y$  is contained in an affine open  $W$  of  $Y$ . There then exists an integer  $n$  and a closed  $W$ -immersion

$$V \longrightarrow W[T_1, \dots, T_n],$$

hence a  $Y$ -immersion  $V \rightarrow Y[T_1, \dots, T_n]$ , which is regular by VII 1.10. Since the composite of two regular immersions is regular, one obtains a regular  $Y$ -immersion

$$U \longrightarrow Y[T_1, \dots, T_n]$$

factorizing  $f$ .

**Proposition 1.2.** Let  $f : X \rightarrow Y$  be a complete-intersection morphism. Then every  $Y$ -immersion  $i : X \rightarrow Z$  of  $X$  in a smooth  $Y$ -scheme is regular.

We first prove the following corollary.

**Corollary 1.3.** Let

$$\begin{array}{ccc} & & V' \\ & \nearrow j & \downarrow p \\ X & \xrightarrow{i} & V \end{array}$$

be a commutative diagram in which  $i$  and  $j$  are immersions, and  $p$  is a smooth morphism. Then  $i$  is a regular immersion if and only if  $j$  is a regular immersion.

(i) Suppose  $i$  regular. Put  $X' = V' \times_V X$ . The datum of  $j$  defines a section  $j'$  of  $X'$  over  $X$ , such that  $j = i'j'$ :

$$\begin{array}{ccc} X' & \xrightarrow{i'} & V' \\ p' \downarrow & \nearrow j & \downarrow p \\ X & \xrightarrow{i} & V. \end{array}$$

Now  $j'$  is a regular immersion, being a section of a smooth morphism (VII 1.10);  $i'$  is a regular immersion because it is obtained from  $i$  by the smooth, hence flat, base change (VII 1.5). Therefore  $j$  is a regular immersion (VII 1.7).

(ii) Suppose  $j$  regular and  $p$  étale. Then  $j'$  is an open immersion (EGA IV 17.9.3), hence an isomorphism of  $X$  onto  $j'(X)$ ; put  $U = j'(X)$ . Let  $i''$  be the restriction of  $i'$  to  $U$ , and let  $U'$  be an open subset of  $V'$  such that  $U = U' \cap X'$ . Then

$$i'' = j \circ (j')^{-1},$$

so that  $i''$  is a regular immersion of  $U$  into  $U'$ . But the morphism  $U' \rightarrow V$ , being smooth, is flat and locally of finite presentation, hence open. Consequently it is faithfully flat and quasi-compact from  $U'$  onto an open neighbourhood of  $X$  in  $V$ . It follows from VII 1.5 that  $i$  is a regular immersion.

(iii) Suppose  $j$  regular, and suppose that  $V'$  is a vector bundle over  $V$ . Since the property to be proved is local on  $V$ , one may suppose  $V$  affine,  $i$  closed, and  $V' = V[T_1, \dots, T_n]$ . Put  $V = \text{Spec}(A)$ ,  $X = \text{Spec}(B)$ . One has the commutative diagram

$$\begin{array}{ccc} & & A[T_1, \dots, T_n] \\ & \nearrow u & \downarrow v \\ A & \longrightarrow & B, \end{array}$$

where the morphisms  $u$  and  $v$  are surjective. Choose elements  $t_i \in A$ ,  $1 \leq i \leq n$ , such that  $u(t_i) = v(T_i)$  for every  $i$ . Define a homomorphism

$$w : A[T_1, \dots, T_n] \longrightarrow A, \quad w(T_i) = t_i.$$

This gives a closed immersion  $s : V \rightarrow V'$  which is a section of  $p$ , and such that  $j = s \circ i$ . Moreover, being a section of a smooth morphism,  $s$  is a regular immersion (VII 1.10). Since  $s$  and  $s \circ i$  are regular, it will follow that  $i$  is regular if one proves that the sequence

$$0 \longrightarrow i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow I/I^2 \longrightarrow 0$$

is exact and locally split; here  $I, J, K$  denote respectively the ideals defining  $i, j, s$  (VII 1.7). It is enough to show that  $i^*(K/K^2)$  is a direct factor of  $J/J^2$ . But  $j'$  is obtained from  $s$  by the base change  $i' : X' \rightarrow V'$ . Moreover  $j'$  and  $s$  are regular immersions (VII 1.10). Thus, by VII 1.2, the ideal defining  $j'$  is  $i'^*(K)$ , and the conormal sheaf of  $j'$  is  $i'^*(K/K^2)$ . The exact sequence of conormal sheaves attached to the decomposition  $j = i'j'$  therefore gives in particular a homomorphism

$$J/J^2 \longrightarrow i^*(K/K^2).$$

It is immediate to verify that the composite

$$i^*(K/K^2) \longrightarrow J/J^2 \longrightarrow i^*(K/K^2)$$

is the identity. In affine terms, this corresponds to the decomposition of  $J$  as the direct sum of an ideal of  $A$  and the ideal generated by the  $T_i - t_i$ . This gives the required splitting.

(iv) Finally suppose  $j$  regular and  $p$  smooth. For every  $x \in X$ , one can find an open neighbourhood  $W$  of  $i(x)$ , an open neighbourhood  $W'$  of  $j(x)$  lying over  $W$ , and a commutative diagram

$$\begin{array}{ccc} W' & \xrightarrow{q} & W[T_1, \dots, T_n] \\ p \downarrow & & \downarrow p_0 \\ W & \xlongequal{\quad} & W, \end{array}$$

where  $p_0$  is the structural morphism, and  $q$  is an étale morphism (EGA IV 17.11.4). One may then suppose  $V = W$  and  $V' = W'$ . It is easy to see that  $qj$  is an immersion, by introducing the fiber product  $X[T_1, \dots, T_n]$ . Applying (ii) and (iii), one deduces that  $qj$  is regular, and then that  $i$  is regular.

We now prove Proposition 1.2. Since regularity is local on  $Z$ , we may work in a neighbourhood of a point  $x \in X$ . There is then an open neighbourhood  $U$  of  $x$  in  $X$ , a smooth  $Y$ -scheme  $V$ , and a regular  $Y$ -immersion  $i' : U \rightarrow V$ . We have a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{i'} & V \\ \downarrow i & & \downarrow \\ Z & \longrightarrow & Y, \end{array}$$

where the unlabeled vertical arrows are the structural morphisms. From it one deduces a closed immersion

$$i'' : U \longrightarrow Z \times_Y V,$$

and a commutative diagram

$$\begin{array}{ccccc} & & U & & \\ & \swarrow i & & \searrow i' & \\ Z & \xleftarrow{p} & Z \times_Y V & \xrightarrow{p'} & V. \end{array}$$

Since  $i'$  is regular and  $p, p'$  are smooth, the corollary implies that  $i''$  and  $i$  are regular.

The following proposition will not be used in the sequel of the Seminar.

**Proposition 1.4.** Let  $f : X \rightarrow Y$  be a flat morphism locally of finite presentation. For  $f$  to be a complete-intersection morphism in the sense of 1.1, it is necessary and sufficient that it be a complete-intersection morphism in the sense of EGA IV 19.3.6.

Suppose that  $f$  is a complete-intersection morphism in the sense of 1.1, and let  $x \in X$ . Then there are an open neighbourhood  $U$  of  $x$  and a factorization

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow f & \downarrow g \\ & & Y, \end{array}$$

where  $i$  is a regular immersion and  $g$  is smooth. Let  $y = f(x)$ . Then the immersion  $i_y : U_y \rightarrow V_y$  is regular at  $x$ . Indeed, after possibly restricting  $V$ , one may suppose that there is a finite-type free  $\mathcal{O}_V$ -module  $E$  and a regular homomorphism  $u : E \rightarrow \mathcal{O}_V$  such that the Koszul complex associated to  $u$  is a resolution of  $\mathcal{O}_U$ . Let  $u_y$  be the homomorphism induced on the fiber over  $y$ , and let  $K_\bullet(u_y) = K_\bullet(u)_y$  be the associated Koszul complex. Since  $g$  is smooth, hence flat,  $K_\bullet(u)$  is a  $Y$ -flat resolution of  $\mathcal{O}_U$ . Consequently the homology groups of  $K_\bullet(u)_y$  are the groups

$$\mathrm{Tor}_i^{\mathcal{O}_Y}(\mathcal{O}_U, k(y)),$$

where  $k(y)$  is the residue field of  $\mathcal{O}_{Y,y}$ . Since  $f$  is flat, these groups are zero; hence  $i_y$  is regular at  $x$ . Since  $g$  is smooth, the local ring of  $V_y$  at  $x$  is regular. The local ring of  $U_y$  at  $y$  is therefore the quotient of a regular ring by a regular ideal, hence a complete-intersection ring (EGA IV 19.3.2). It follows that  $f$  is a complete intersection at  $x$  in the sense of EGA IV 19.3.6.

Conversely, suppose that  $f$  is a complete-intersection morphism in the sense of EGA IV 19.3.6, and let  $x \in X$ . Let  $U$  be an affine neighbourhood of  $x$  lying over an affine neighbourhood  $U'$  of  $y = f(x)$ . One can then find a smooth  $U'$ -scheme  $V$ , and a closed immersion  $U \rightarrow V$  (for instance by taking for  $V$  a vector bundle over  $U'$ ). We shall show that  $i$  is regular. By EGA IV 19.3.7,  $i$  is transversally regular at  $x$ , since  $V$  is smooth over  $Y$ , hence  $V_y$  is regular at  $x$ , and  $U$  is flat over  $Y$ . It follows from EGA IV 19.2.4, a)3c), that  $i$  is regular at  $x$ . Therefore  $f$  is a complete-intersection morphism in the sense of 1.1.

**Proposition 1.5.** Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be two locally complete-intersection morphisms. Then  $gf$  is locally complete-intersection.

Let  $x \in X$ . One can find open neighbourhoods  $U, V$  of  $x, f(x)$  and regular immersions  $i, j$  such that one has factorizations

$$\begin{array}{ccc} U & \xrightarrow{i} & V[T_1, \dots, T_n] \\ & \searrow f & \downarrow \\ & & V \end{array} \quad \begin{array}{ccc} V & \xrightarrow{j} & Z[T'_1, \dots, T'_m] \\ & \searrow g & \downarrow \\ & & Z. \end{array}$$

This diagram can be completed by the commutative square

$$\begin{array}{ccc} V[T_1, \dots, T_n] & \xrightarrow{j'} & Z[T'_1, \dots, T'_m, T_1, \dots, T_n] \\ \downarrow & & \downarrow \\ V & \xrightarrow{j} & Z[T'_1, \dots, T'_m], \end{array}$$

where  $j'$  is obtained from  $j$  by base change. This base change is flat; hence  $j'$  is a regular immersion, and so  $j'i$  is a regular immersion of  $U$  in a scheme smooth over  $W$ , where  $W$  is the relevant open of  $Z$ . This proves the assertion.

**Proposition 1.6.** Let  $f : X \rightarrow Y$  be a morphism and let  $g : Y' \rightarrow Y$  be a flat base change. If  $f$  is complete-intersection, then the morphism

$$f' : X \times_Y Y' \rightarrow Y'$$

deduced from  $f$  is complete-intersection. The converse is true if, moreover,  $g$  is quasi-compact and surjective.

One can find an open covering of  $X$  by open subsets  $U_i$  such that, for every  $i$ , the restriction of  $f$  to  $U_i$  factors as

$$U_i \rightarrow V_i \rightarrow Y,$$

where  $U_i \rightarrow V_i$  is a regular immersion and  $V_i \rightarrow Y$  is smooth. The  $U_i \times_Y Y'$  form an open covering of  $X \times_Y Y'$ , the immersions

$$U_i \times_Y Y' \rightarrow V_i \times_Y Y'$$

are regular by VII 1.5, and the morphisms  $V_i \times_Y Y' \rightarrow Y'$  are smooth, giving the first assertion. For the converse, EGA IV 2.7.1 (iv) first implies that  $f$  is locally of finite presentation; this reduces us easily, using 1.2, to the case where  $f$  is a closed immersion. In that case the assertion is just VII 1.5.

**Proposition 1.7.** Let  $f : X \rightarrow Y$  be a complete-intersection morphism. Then  $f$  is a perfect morphism.

Since the property is local on  $X$ , one may work in a neighbourhood of a point  $x \in X$  and suppose that there is an open subset  $U$  containing  $x$  and a factorization

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow f & \downarrow g \\ & & Y, \end{array}$$

where  $i$  is a regular immersion and  $g$  is smooth. Since  $g$  is smooth, it is a perfect morphism (III 4.1). By VII 1.9,  $i$  is also a perfect morphism. Since the composite of two perfect morphisms is perfect (III 4.5), the proposition follows.

**Proposition 1.8.** Let  $f : X \rightarrow Y$  be a complete-intersection morphism, let  $x$  be a point of  $X$ , and let

$$\begin{array}{ccc} U & \xrightarrow{i} & V \\ & \searrow f & \downarrow g \\ & & Y \end{array}$$

be a factorization in which  $U$  is an open neighbourhood of  $x$ ,  $i$  is a regular immersion, and  $g$  is smooth. Let  $J$  be the ideal of  $\mathcal{O}_V$  defining  $U$ . Then the integer

$$d_x = \text{rank}((\Omega_{V/Y}^1)_{i(x)}) - \text{rank}((J/J^2)_x)$$

is independent of the chosen factorization of  $f$ .

Suppose that the restriction of  $f$  to  $U$  factors through two smooth  $Y$ -schemes  $V$  and  $V'$ , and put  $V'' = V \times_Y V'$ . Then  $V''$  is smooth over  $Y$ , and also over  $V$  and  $V'$ . Moreover, the immersions  $X \rightarrow V$  and  $X \rightarrow V'$  define an immersion  $X \rightarrow V''$ , which is regular by 1.2. It is therefore enough to compare the integers  $d_x$  attached to two factorizations through smooth  $Y$ -schemes  $V$  and  $V'$  such that there exists a commutative diagram over  $Y$

$$\begin{array}{ccc} & & V' \\ & \nearrow j & \downarrow p \\ X & \xrightarrow{i} & V, \end{array}$$

where  $p$  is smooth. We may also suppose that  $\Omega_{V/Y}^1$ ,  $\Omega_{V'/Y}^1$ , and  $\Omega_{V'/V}^1$  have constant rank, as do the conormal sheaves of  $i$  and  $j$ .

From the commutative diagram, all morphisms being smooth,

$$\begin{array}{ccc} V' & \xrightarrow{p} & V \\ & \searrow & \downarrow \\ & & Y, \end{array}$$

one deduces

$$\text{rank}(\Omega_{V'/Y}^1) = \text{rank}(\Omega_{V/Y}^1) + \text{rank}(\Omega_{V'/V}^1). \quad (1.8.1)$$

Let  $X' = V' \times_V X$ ; then the rank of  $\Omega_{V'/V}^1$  is equal to that of  $\Omega_{X'/X}^1$ . Further,  $j$  defines a section  $j' : X \rightarrow X'$ , which is a regular immersion of codimension equal to the rank of  $\Omega_{X'/X}^1$  (EGA IV

17.10.4). If  $i'$  is the immersion obtained from  $i$  by the base change  $V' \rightarrow V$ , then  $j = i'j'$ , and consequently (VII 1.7)

$$\mathrm{rank}(N_{V'/X}) = \mathrm{rank}(N_{V/X}) + \mathrm{rank}(\Omega_{V'/V}^1). \quad (1.8.2)$$

Comparing (1.8.1) and (1.8.2) gives the proposition.

**Definition 1.9.** The integer  $d_x$  of Proposition 1.8 is called the *relative virtual dimension* of  $X$  over  $Y$  (or of  $f$ ) at the point  $x$ .

The relative virtual dimension of  $X$  over  $Y$  is therefore a locally constant function on  $X$ , with integral values, positive or negative. If  $f$  is a smooth morphism of relative dimension  $n$ , its relative virtual dimension is  $n$ . If  $f$  is a regular immersion of codimension  $n$ , its relative virtual dimension is  $-n$ .

**Proposition 1.10.** Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be two complete-intersection morphisms, and let  $x \in X$ . Then

$$\dim. \mathrm{rel.} \mathrm{virt.}_x(gf) = \dim. \mathrm{rel.} \mathrm{virt.}_x(f) + \dim. \mathrm{rel.} \mathrm{virt.}_{f(x)}(g).$$

This follows immediately from 1.9 and from the proof of 1.5.

## 2. The relative cotangent complex

**2.1.** We give here a definition of the cotangent complex which, without having the greatest possible generality (see Exposé 0, 4.4), will be sufficient for the needs of the Riemann–Roch theorem. Of course the definition given in Exposé 0 coincides, in the case considered there, with the definition given here.

Let  $f : X \rightarrow Y$  be a smoothable morphism, that is, a morphism admitting a factorization

$$\begin{array}{ccc} X & \xrightarrow{i} & V \\ & \searrow f & \downarrow g \\ & & Y, \end{array}$$

where  $g$  is smooth and  $i$  is an immersion, which may be supposed closed. Let  $J$  be the ideal of  $X$  in  $V$ . The canonical derivation

$$d : \mathcal{O}_V \longrightarrow \Omega_{V/Y}^1$$

defines a homomorphism

$$J \longrightarrow \mathcal{O}_V \xrightarrow{d} \Omega_{V/Y}^1 \longrightarrow \Omega_{V/Y}^1 \otimes_{\mathcal{O}_V} \mathcal{O}_X$$

which vanishes on  $J^2$ , and hence defines a homomorphism, still denoted  $d$ ,

$$J/J^2 \xrightarrow{d} i^*(\Omega_{V/Y}^1). \quad (2.1.1)$$

One checks easily that this homomorphism is  $\mathcal{O}_X$ -linear. One then defines a chain complex  $L_{\bullet}^{X/Y}$  by putting

$$L_0^{X/Y} = i^*(\Omega_{V/Y}^1), \quad L_1^{X/Y} = J/J^2, \quad L_i^{X/Y} = 0 \quad (i \neq 0, 1),$$

and taking as homomorphism  $L_1^{X/Y} \rightarrow L_0^{X/Y}$  the homomorphism (2.1.1).

**Proposition 2.2.** The complex  $L_{\bullet}^{X/Y}$  is independent, up to canonical isomorphism in the derived category  $D(X)$ , of the factorization chosen through a smooth  $Y$ -scheme.

Let  $V, V'$  be two smooth  $Y$ -schemes through which  $f$  factors. Put  $V'' = V \times_Y V'$ . Then  $V''$  is smooth over  $Y$ , and also over  $V$  and  $V'$ . Moreover the immersions  $X \rightarrow V$  and  $X \rightarrow V'$  define an immersion  $X \rightarrow V''$  which factors  $f$ . It is therefore enough to compare the complexes  $L_{\bullet}^{X/Y}$  attached to two factorizations through smooth  $Y$ -schemes  $V$  and  $V'$  for which there is a commutative diagram over  $Y$

$$\begin{array}{ccc} & & V' \\ & \nearrow j & \downarrow p \\ X & \xrightarrow{i} & V, \end{array}$$

where  $i$  and  $j$  are closed immersions, and  $p$  is smooth. Functoriality then gives homomorphisms

$$I/I^2 \longrightarrow J/J^2, \quad p^*(\Omega_{V/Y}^1) \longrightarrow \Omega_{V'/Y}^1,$$

hence a diagram

$$\begin{array}{ccc} I/I^2 & \longrightarrow & J/J^2 \\ d \downarrow & & \downarrow d \\ i^*(\Omega_{V/Y}^1) & \longrightarrow & j^*(\Omega_{V'/Y}^1). \end{array} \quad (2.2.1)$$

Since  $V$  and  $V'$  are smooth over  $Y$ , one obtains the exact sequence

$$0 \longrightarrow p^*(\Omega_{V/Y}^1) \longrightarrow \Omega_{V'/Y}^1 \longrightarrow \Omega_{V'/V}^1 \longrightarrow 0.$$

Since the modules of differentials considered are locally free on  $\mathcal{O}_V$ , applying the functor  $j^*$  gives the exact sequence

$$0 \longrightarrow i^*(\Omega_{V/Y}^1) \longrightarrow j^*(\Omega_{V'/Y}^1) \longrightarrow j^*(\Omega_{V'/V}^1) \longrightarrow 0. \quad (2.2.2)$$

Furthermore,  $j$  defines a section  $j'$  of  $X' = V' \times_V X$ :

$$\begin{array}{ccc} X' & \xrightarrow{i'} & V' \\ p' \downarrow & \nearrow j & \downarrow p \\ X & \xrightarrow{i} & V. \end{array}$$

Let  $I, J, K$  be the ideals defining  $i, j, j'$ . Then  $j = i'j'$ , and the immersions  $i', j, j'$  are regular; moreover  $i'$  is defined by the ideal  $p^*(I)$ , because  $p$  is flat, and its conormal sheaf is therefore

$$i'^*(p^*(I)) = p'^*(I/I^2).$$

One obtains the exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow J/J^2 \longrightarrow K/K^2 \longrightarrow 0. \quad (2.2.3)$$

We shall show: (a) the diagram (2.2.1) is commutative; this will give a homomorphism of complexes  $\theta$  from the complex  $L_{\bullet}^{X/Y}$  defined by  $V$  to the complex  $L_{\bullet}^{X/Y}$  defined by  $V'$ ; (b) the homomorphism

$$K/K^2 \longrightarrow j^*(\Omega_{V'/V}^1)$$

defined by (2.2.2), (2.2.3), and (2.2.1) is an isomorphism. This will imply that the mapping cylinder of  $\theta$  is acyclic, hence that  $\theta$  is a quasi-isomorphism, and therefore defines an isomorphism in the derived category.

The properties (a) and (b) are local on  $X$ . One may therefore suppose  $X, Y, V, V'$  affine. Put  $Y = \text{Spec}(A)$ ,  $V = \text{Spec}(B)$ ,  $V' = \text{Spec}(B')$ ,  $X = \text{Spec}(C)$ , and keep the same letters for the corresponding homomorphisms and ideals. One has the diagram

$$\begin{array}{ccccc}
 B \otimes_A B & \xrightarrow{p \otimes p} & B' \otimes_A B' & \xleftarrow{\quad} & B' \otimes_B B' \\
 \Delta \downarrow & & \downarrow \Delta & & \downarrow \\
 B & \xrightarrow{p} & B' & & B' \otimes_B C = B'/IB' \\
 & \searrow i & \downarrow j & \swarrow p' & \\
 & & C & & 
 \end{array}$$

where the two displayed squares are those denoted (1) and (2) in the source. The homomorphism  $I/I^2 \rightarrow J/J^2$  is deduced from the homomorphism  $IB' \rightarrow J$  and coincides with that deduced from  $p : I \rightarrow J$ ; the homomorphism

$$i^*(\Omega_{V/Y}^1) \longrightarrow j^*(\Omega_{V'/Y}^1)$$

is deduced from the commutative square (1). Hence property (a) is evident.

The homomorphism  $K/K^2 \rightarrow j^*(\Omega_{V'/Y}^1)$  considered in (b) may be defined as follows. Let  $\bar{x} \in K/K^2$ , and let  $x$  be an element of  $J$  lifting  $\bar{x}$  (such an element exists because  $K = J/IB'$ ). To  $\bar{x}$  one associates the image of  $x$  by the composite

$$J \xrightarrow{d} \Omega_{B'/A}^1 \otimes_{B'} C \longrightarrow \Omega_{B'/B}^1 \otimes_{B'} C.$$

On the other hand, let  $H$  be the augmentation ideal of

$$B' \otimes_B B' \longrightarrow B'.$$

The commutative square (2) gives a homomorphism  $H \rightarrow K$ , hence a homomorphism

$$H/H^2 \otimes_{B'} C \longrightarrow K/K^2.$$

But  $H/H^2 = \Omega_{B'/B}^1$ . It remains only to check that the homomorphisms thus constructed are inverse to each other, which is straightforward.

**Definition 2.3.** Let  $f : X \rightarrow Y$  be a smoothable morphism. The complex  $L_{\bullet}^{X/Y}$ , defined up to canonical isomorphism in the derived category  $D(X)$ , is called the *relative cotangent complex* of  $X$  over  $Y$  (or of  $f$ ).

**Proposition 2.4.** Let  $f : X \rightarrow Y$  be a smoothable complete-intersection morphism. Then  $L_{\bullet}^{X/Y}$  is perfect. More precisely, for every factorization of  $f$  as the composite of an immersion and a smooth morphism, the complex  $L_{\bullet}^{X/Y}$  defined in 2.1 is strictly perfect.

Let  $f = gi$ , where  $g$  is smooth and  $i$  is an immersion. By 1.2,  $i$  is a regular immersion. If  $J$  is the ideal defining  $i$ , then  $J/J^2$  is a locally free finite-type  $\mathcal{O}_X$ -module. Likewise, since  $g : V \rightarrow Y$  is smooth,  $i^*(\Omega_{V/Y}^1)$  is locally free of finite type. This proves 2.4.

**Corollary 2.5.** Let  $f : X \rightarrow Y$  be a smoothable complete-intersection morphism, and let  $f = gi$  be a factorization of  $f$  through a smooth  $Y$ -scheme  $V$ . Let  $J$  be the ideal of  $\mathcal{O}_V$  defining  $i$ . If



$K^\bullet(X)$  denotes the Grothendieck group of locally free finite-type  $\mathcal{O}_X$ -modules (respectively of perfect complexes on  $X$ ), the element  $T_f \in K^\bullet(X)$  defined by

$$T_f = \text{cl}(i^* \Omega_{V/Y}^1) - \text{cl}(J/J^2)$$

respectively

$$T_f = \text{cl}(L_\bullet^{X/Y})$$

does not depend on the chosen factorization of  $f$ . The value at a point  $x$  of  $X$  of the augmentation of  $T_f$  is precisely the relative virtual dimension of  $X$  over  $Y$  at  $x$  (1.9).

The element  $T_f$  is well defined in both cases by 2.4; its class in  $K^\bullet(X)$  is independent of the factorization, by 2.2 in the second case and by the proof of 2.2 in the first. The last assertion is trivial.

**Proposition 2.6.** Let  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  be two smoothable complete-intersection morphisms, and suppose in addition that the following condition is satisfied:

(G) There exists a smooth  $Y$ -scheme  $V$  factorizing  $f$  such that, for every closed immersion  $Y \rightarrow V'$  into a smooth  $Z$ -scheme, one can find a smooth  $V'$ -scheme  $V''$  with

$$V \simeq V'' \times_{V'} Y.$$

This condition is in particular satisfied if  $f$  is quasi-projective and if  $Y$  has an ample sheaf, by taking for  $V$  some  $\mathbb{P}_Y^r$ .

Then  $gf$  is complete-intersection and smoothable, and there is a distinguished triangle

$$\begin{array}{ccc} & L_\bullet^{X/Y} & \\ \swarrow & & \searrow \\ f^* L_\bullet^{Y/Z} & \xrightarrow{\quad} & L_\bullet^{X/Z} \end{array}$$

in the derived category.

Let  $V'$  be a smooth  $Z$ -scheme factorizing  $g$ . Consider the commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{i} & V & \xrightarrow{j'} & V'' \\ & \searrow f & \downarrow p & & \downarrow p'' \\ & & Y & \xrightarrow{j} & V' \\ & & & \searrow g & \downarrow p' \\ & & & & Z. \end{array}$$

Then the immersion  $j'$  is regular, since it is obtained from  $j$  by a flat base change. Hence  $j'i$  is a regular immersion factorizing  $gf$ , which is therefore smoothable and complete-intersection. Moreover, one has the exact sequence of conormal sheaves

$$0 \longrightarrow i^*(J'/J'^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0,$$

where  $I, J, J', K$  denote the sheaves defining respectively the immersions  $i, j, j', j'i$ . Since  $p''$  is flat,  $J' = p''^*(J)$ , and the sequence can be written

$$0 \longrightarrow f^*(J/J^2) \longrightarrow K/K^2 \longrightarrow I/I^2 \longrightarrow 0. \quad (2.6.1)$$

Furthermore, since  $V'$  and  $V''$  are smooth over  $Z$ , there is an exact sequence

$$0 \longrightarrow p''^*(\Omega_{V'/Z}^1) \longrightarrow \Omega_{V''/Z}^1 \longrightarrow \Omega_{V''/V'}^1 \longrightarrow 0.$$

Applying  $(j'i)^*$ , one obtains an exact sequence

$$0 \longrightarrow (p''ji)^*(\Omega_{V'/Z}^1) \longrightarrow (j'i)^*(\Omega_{V''/Z}^1) \longrightarrow (j'i)^*(\Omega_{V''/V'}^1) \longrightarrow 0.$$

Since

$$j'^*(\Omega_{V''/V'}^1) = \Omega_{V/Y}^1$$

by hypothesis (G), one obtains

$$0 \longrightarrow f^*(j^*\Omega_{V'/Z}^1) \longrightarrow (j'i)^*(\Omega_{V''/Z}^1) \longrightarrow i^*(\Omega_{V/Y}^1) \longrightarrow 0. \quad (2.6.2)$$

Combining (2.6.1) and (2.6.2), one is reduced to checking that the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & f^*(J/J^2) & \longrightarrow & K/K^2 & \longrightarrow & I/I^2 \longrightarrow 0 \\ & & \downarrow f^*(d) & & \downarrow d & & \downarrow d \\ 0 & \longrightarrow & f^*(j^*\Omega_{V'/Z}^1) & \longrightarrow & (j'i)^*\Omega_{V''/Z}^1 & \longrightarrow & i^*\Omega_{V/Y}^1 \longrightarrow 0 \end{array}$$

is commutative. This is verified locally without difficulty by returning to the definitions of the homomorphisms involved. The announced distinguished triangle follows from the resulting exact sequence of complexes.

**Corollary 2.7.** With the notation and hypotheses of 2.6, one has in  $K^\bullet(X)$  the equality

$$T_{gf} = T_f + f^*(T_g).$$

### 3. Riemann–Roch theorem: statement

**3.1.** In no. 3,  $Y$  is a quasi-compact scheme having an ample sheaf.

Let  $f : X \rightarrow Y$  be a projective complete-intersection morphism (1.1); suppose moreover that the relative virtual dimension of  $f$  (1.9) is constant on  $X$ . Since  $f$  is projective, there is a closed immersion  $X \rightarrow P$ , where  $P$  is a projective bundle over  $Y$ ; since  $Y$  has an ample sheaf, one may suppose that  $P = \mathbb{P}_Y^r$ . Thus there is a factorization

$$\begin{array}{ccc} X & \xrightarrow{i} & \mathbb{P}_Y^r \\ & \searrow f & \downarrow g \\ & & Y. \end{array}$$

Since  $f$  is complete-intersection, the immersion  $i$  is regular (1.2).

By 1.7,  $f$  is a perfect morphism. Moreover  $Y$  has an ample sheaf, and therefore so do  $X$  and  $P$ . One can therefore define an additive homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y)$$

(IV 2.12), the groups  $K^\bullet$  defined in terms of locally free sheaves of finite type, or of perfect complexes, being identical. One can also define additive homomorphisms

$$i_* : K^\bullet(X) \longrightarrow K^\bullet(P), \quad g_* : K^\bullet(P) \longrightarrow K^\bullet(Y),$$

and one has  $f_* = g_* i_*$ .

Let  $d$  be the relative virtual dimension of  $X$  over  $Y$ . The codimension of  $i$  is then  $r - d$ . By VII 4.6, the homomorphism  $i_*$  gives, after tensoring with  $\mathbb{Q}$ , inclusions

$$i_*(\mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}) \subset \mathrm{Fil}^{k+r-d} K^\bullet(P)_{\mathbb{Q}}.$$

By VI 5.8, the homomorphism  $g_*$  gives inclusions

$$g_*(\mathrm{Fil}^{k'} K^\bullet(P)) \subset \mathrm{Fil}^{k'-r} K^\bullet(Y).$$

Composing, one obtains the following.

**Proposition 3.2.** Let  $f : X \rightarrow Y$  be a projective complete-intersection morphism whose relative virtual dimension  $d$  is constant. Suppose that  $Y$  is quasi-compact and has an ample sheaf. Then for every  $k \in \mathbb{Z}$ ,

$$f_*(\mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}) \subset \mathrm{Fil}^{k-d} K^\bullet(Y)_{\mathbb{Q}}.$$

Consequently one obtains an additive homomorphism

$$f_*^{\mathrm{gr}} : \mathrm{Gr} K^\bullet(X)_{\mathbb{Q}} \longrightarrow \mathrm{Gr} K^\bullet(Y)_{\mathbb{Q}}$$

sending  $\mathrm{Gr}^k K^\bullet(X)_{\mathbb{Q}}$  into  $\mathrm{Gr}^{k-d} K^\bullet(Y)_{\mathbb{Q}}$ .

**3.3.** Recall that in V 6.1 we defined a functor

$$A \longmapsto \mathrm{Ch}(A)$$

from the category of graded  $K$ -algebras to the category of augmented  $K$ - $\lambda$ -algebras, where  $K$  is a binomial ring (V 2.7). If  $A$  is a graded  $K$ -algebra, then

$$\mathrm{Ch}(A) = K \times (1 + \hat{A}^+).$$

For every augmented  $K$ - $\lambda$ -algebra  $\Lambda$  we defined a  $K$ - $\lambda$ -homomorphism, called the completed Chern class (V 6.7),

$$\tilde{c} = (\varepsilon, c) : \Lambda \longrightarrow \mathrm{Ch}(\mathrm{Gr}_\Lambda).$$

Finally, we defined a ring homomorphism, functorial in  $A$ , called the Chern character (V 6.4),

$$\mathrm{ch} : \mathrm{Ch}(A) \longrightarrow K \oplus \hat{A}_{\mathbb{Q}}^+,$$

and a group endomorphism, called the Todd character (V 1.16),

$$\mathrm{Todd} : 1 + \hat{A}^+ \longrightarrow 1 + \hat{A}^+.$$

We shall apply these definitions to the augmented  $\lambda$ -rings  $K^\bullet(X)$  and  $K^\bullet(Y)$ ; the augmentation rings  $H^0(X, \mathbb{Z})$  and  $H^0(Y, \mathbb{Z})$  are binomial (V 2.8). To simplify notation, for every scheme  $Z$  and every  $z \in K^\bullet(Z)$  we write

$$\mathrm{ch}(z) = \mathrm{ch}(\tilde{c}(z)), \quad \mathrm{Todd}(z) = \mathrm{Todd}(c(z)).$$

For every graded algebra  $A$  we regard the group  $1 + \hat{A}^+$  as contained in  $\hat{A}$ . Finally, if  $f : A \rightarrow B$  is additive and compatible with the gradings up to a shift, we shall still write  $f$  for the homomorphism  $\hat{A} \rightarrow \hat{B}$  deduced from it.

**3.4.** Let  $X$  be a scheme. For every locally free finite-type  $\mathcal{O}_X$ -module  $E$ , denote by  $\check{E}$  the class in  $K^\bullet(X)$  of the dual  $\check{E}$  of  $E$ . It is clear that the function  $E \mapsto \check{E}$  is additive on the category of locally free finite-type  $\mathcal{O}_X$ -modules, and therefore defines an endomorphism of  $K^\bullet(X)$ , denoted  $x \mapsto \check{x}$ .

**Proposition 3.5.** Let  $X$  be a quasi-compact scheme. Then, for all  $x \in K^\bullet(X)$  and all  $i \geq 0$ , one has

$$c^i(\check{x}) = (-1)^i c^i(x).$$

Equivalently,

$$c_t(\check{x}) = c_{-t}(x), \quad c_t(x) = \sum_{i \geq 0} c^i(x) t^i.$$

By additivity one may suppose that  $x$  is the class of a locally free finite-type  $\mathcal{O}_X$ -module  $E$ . Moreover, after replacing  $X$  by the flag scheme of  $E$ , which gives an injection on the groups  $\mathrm{Gr} K^\bullet$  (VI 5.5), one may suppose that  $E$  is an invertible sheaf. Then  $\check{E} = E^{-1}$ , so  $\check{x} = x^{-1}$ .

The  $c^i(\check{x})$  and  $c^j(x)$  are zero for  $i, j \geq 2$ . It is enough to prove

$$c^1(x^{-1}) = -c^1(x).$$

Now  $c^1(x)$  is the class in  $\mathrm{Gr}^1 K^\bullet(X)$  of  $x - 1$ , while  $c^1(x^{-1})$  is that of  $x^{-1} - 1$ . The equality

$$x^{-1} - 1 = 1 - x + (x - 1)(1 - x^{-1})$$

gives

$$x^{-1} - 1 = 1 - x \pmod{\mathrm{Fil}^2 K^\bullet(X)},$$

and the result follows.

**Theorem 3.6 (Riemann–Roch).** Let  $f : X \rightarrow Y$  be a projective complete-intersection morphism (1.1), whose relative virtual dimension  $d$  is constant (1.9). Suppose that  $Y$  is quasi-compact and admits an ample sheaf. With the notation of 3.3, for every  $x \in K^\bullet(X)$  one has

$$\mathrm{ch}(f_*(x)) = f_*^{\mathrm{gr}}(\mathrm{ch}(x) \mathrm{Todd}(\check{T}_f)),$$

where  $\check{T}_f$  denotes the dual element of the class of the relative cotangent complex  $L_{\bullet}^{X/Y}$  (2.5 and 2.3).

In other words, the diagram

$$\begin{array}{ccc} K^\bullet(X) & \xrightarrow{\mathrm{ch} \circ \check{c}} & \mathrm{Gr} K^\bullet(X)_{\mathbb{Q}} \\ f_* \downarrow & & \downarrow x \mapsto f_*^{\mathrm{gr}}(x \mathrm{Todd}(\check{T}_f)) \\ K^\bullet(Y) & \xrightarrow{\mathrm{ch} \circ \check{c}} & \mathrm{Gr} K^\bullet(Y)_{\mathbb{Q}} \end{array}$$

is commutative.